Vedantu

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2017 - Mathematics

Maximum Marks: 100

1. $f(x) = x^2 + 5$, then $f(-4) =$

(a) 26

(b) 21

(c) 20

(d) –20

Answer: (b)

Solution: 21

$f(4) = (-4)^2 + 5$

$= 16 + 5 = 21$

2. If k + 2, 4k – 6, k – 2 are the three consecutive terms of an A.P., then the value of k is:

(a) 2

(b) 3

(c) 4

(d) 5

Answer: (b)

Solution: 3

$2b = a + c$

$2(4k - 6) = k + 2 + 3k - 2$

$8k - 12 = 4k$

$4k = 12$

$k = 3$

3. If the product of the first four consecutive terms of a G.P. is 256 and if the common ratio is 4 and the first term is positive, then its 3rd term is:

(a) 8

(b) $\frac{1}{16}$

(c) $\frac{1}{32}$

(d) 16

Answer: (a) 8

Solution: 8

4. The remainder when $x^2 - 2x + 7$ is divided by $x + 4$ is:

(a) 28

(b) 29

(c) 30

(d) 31

Answer: (d)

Solution: 31

$$\begin{array}{l}
x+4\overline{)x^2-2x+7}(x-6 \\
\quad\;\; \cancel{x^2}+4x \\
\quad\;\; \underline{-\quad -\qquad\;} \\
\qquad\quad -6x+7 \\
\qquad\quad -6x-24 \\
\qquad\quad \underline{+\qquad +\;\;} \\
\qquad\qquad\quad 31
\end{array}$$

5. The common root of the equations $x^2 - \mathrm{b}x + \mathrm{C} = 0$ and $x^2 + \mathrm{b}x - \mathrm{a} = 0$ is:

(a) $\frac{\mathrm{c}+a}{2\mathrm{b}}$

(b) $\frac{\mathrm{c}-\mathrm{a}}{2\mathrm{b}}$

(c) $\frac{c+b}{2a}$

(d) $\frac{a+b}{2c}$

Answer: (a)

Solution: $\frac{c+a}{2b}$

6. $A=\begin{pmatrix}7 & 2\\ 1 & 3\end{pmatrix}$ and $A+B=\begin{pmatrix}-1 & 0\\ 2 & -4\end{pmatrix}$, then the matrix B =

(a) $\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}$

(b) $\begin{pmatrix}6 & 2\\ 3 & -1\end{pmatrix}$

(c) $\begin{pmatrix}-8 & -2\\ 1 & -7\end{pmatrix}$

(d) $\begin{pmatrix}8 & 2\\ -1 & 7\end{pmatrix}$

Answer: (c)

Solution: $B=\begin{bmatrix}-1 & 0\\ 2 & -4\end{bmatrix}-\begin{bmatrix}7 & 2\\ 1 & 3\end{bmatrix}$

$$=\begin{bmatrix}-8 & -2\\ +1 & -7\end{bmatrix}$$

7. Slope of the straight line which is perpendicular to the straight line joining the points (–2, 6) and (4, 8) is equal to;

(a) $\frac{1}{3}$

(b) 3

(c) –3

(d) $-\frac{1}{3}$

Answer: (c)

Solution: $\frac{1}{3}$

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{8-6}{4+2} = \frac{\cancel{2}}{\cancel{6}}\ \frac{1}{3}$$

8. If the points (2, 5), (4, 6), (a, a) are collinear, then the value of ‘a’ is equal to:

(a) –8

(b) 4

(c) –4

(d) 8

Answer: (d)

Solution: 8

$$\frac{1}{2}\left[x_1\left(y_2 - y_3\right) + x_2\left(y_3 - y_1\right) + x_3\left(y_1 - y_2\right)\right] = 0$$

$$\frac{1}{2}\left[2\left(6-a\right) + 4\left(a-5\right) + a\left(5-6\right)\right] = 0$$

$$\frac{1}{2}\left[12 - 2a + 4a - 20 - a\right] = 0$$

$$\boxed{a = 8}$$

9. The perimeters of two similar triangles are 24 cm and 18 cm respectively. If one side of the first triangle is 8 cm, then the corresponding side of the other triangle is:

(a) 4 cm

(b) 3 cm

(c) 9 cm

(d) 6 cm

Answer: (d)

Solution: 6 cm

$$\frac{P.of\,\Delta_1}{P.of\,\Delta_2} = \frac{side_1}{side_2}$$

$$\frac{24}{18} = \frac{8}{S_2}$$

$$S_2 = \frac{\cancel{8} \times \cancel{18}^{6}}{\cancel{24}^{8}}$$

10. ΔABC is a right angled triangle where ∠B = 90° and BD⊥AC. If BD = 8 cm, AD = 4 cm, then CD is:

(a) 24 cm

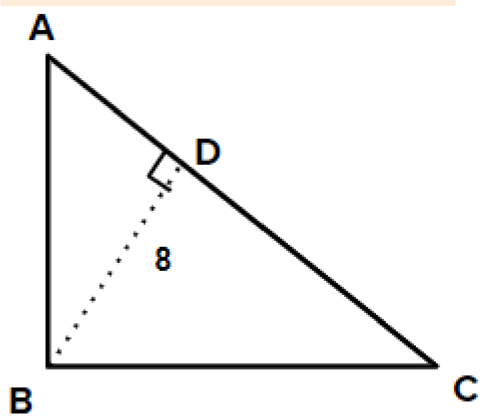
(b) 16 cm

(c) 32 cm

(d) 8 cm

Answer: (b)

Solution: 16 cm



In fig $\Delta ABD \sim \Delta BCD$

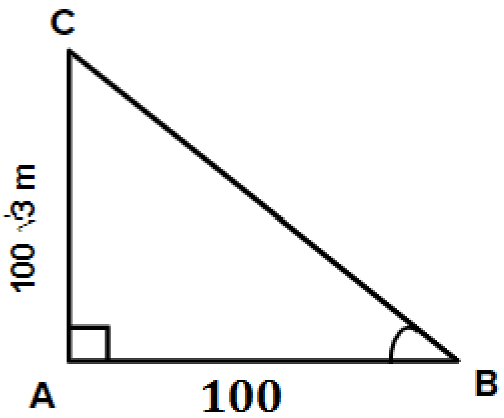
$$\frac{BD}{CD} = \frac{AD}{BD}$$

$$BD^2 = AD \times CD$$

$$8^2 = 4 \times CD$$

$$CD = \frac{64}{4} = 16\,cm$$

11. In the adjoining figure ∠ABC =



(a) 45°

(b) 30°

(c) 60°

(d) 50°

Answer: (c)

Solution: 60°

$$\tan\theta = \frac{100\sqrt{3}}{100}$$

θ = tan 60

12. 9 $\tan^2\theta - 9\sec^2\theta$ =

(a) 1

(b) 0

(c) 9

(d) –9

Answer: (d)

Solution: –9

$9\tan^2\theta - 9\sec^2\theta$

$-9(\sec^2\theta - \tan^2\theta)$

$-9 \times 1 = -9$

13. If the surface area of a sphere is $100\,\pi\ \text{cm}^2$, then is radius is equal to:

(a) 25 cm

(b) 100 cm

(c) 5 cm

(d) 10 cm

Answer: (c)

Solution: 5 cm

$4\cancel{\pi}r^2 = 100\cancel{\pi}$

$r = \sqrt{\dfrac{100}{4}} = 5$

14. Standard deviation of a collection of a data is $2\sqrt{2}$. If each value is multiplied by 3, then the standard deviation of the new data is:

(a) $\sqrt{12}$

(b) $4\sqrt{2}$

(c) $6\sqrt{2}$

(d) $9\sqrt{2}$

Answer: (c)

Solution: $6\sqrt{2}$

15. A card is drawn from a pack of 52 cards at random. The probability of getting neither an ace nor a king card is:

(a) $\dfrac{2}{13}$

(b) $\frac{11}{13}$

(c) $\frac{4}{13}$

(d) $\frac{8}{13}$

Answer: (b)

Solution: $\frac{11}{13}$

16. Given, A = {1, 2, 3, 4, 5} B = {3, 4, 5, 6} and C = {5, 6, 7, 8}, show that $A \cup (B \cup C) = (A \cup B) \cup C$.

Solution:

$A \cup (B \cup C) = \{1, 2, 3, 4, 5, 6, 7, 8\}$

$(A \cup B) \cup C = \{1, 2, 3, 4, 5, 6, 7, 8\}$

17. The following table represents a function from A = {5, 6, 8, 10} to B = {19, 15, 9, 11} where $f(x) = 2x - 1$. Find the values of a and b.

x	5	6	8	10
$f(x)$	a	11	b	19

Solution:

a = 9 b = 15

$Px = 2x - 1$

@ x = 5 $f(x)$ or a = 2 × 5 – 1 = 9

@ x = 8 $f(x)$ or b = 2 × 8 – 1 = 15.

18. If $-\frac{2}{7}, m, -\frac{7}{2}(m+2)$ are in G.P., find the values of m.

Solution:

$$\frac{m}{(-2/7)} = \frac{-7(m+2)/2}{m}$$

$$m^2 = \frac{\cancel{-7}}{\cancel{2}}(m+2) \times \frac{\cancel{-2}}{\cancel{7}}$$

$$m^2 - m - 2 = 0$$

$$m^2 - 2m + m - 2 = 0$$

$$m(m-2) + 1(m-2)$$

$$m = 2,\ \ m = -1$$

19. Solve by elimination method: $13x + 11y = 70$, $11x + 13y = 74$.

Solution:

$13x + 11y = 70 \times 13$

$11x + 13y = 74 \times 11$

$$169x + 14\cancel{3}y = 910 \qquad \ldots(1)$$

$$\underset{-}{121x} \underset{-}{+} 14\cancel{3}y = \underset{-}{814} \qquad \ldots(2)$$

$48x = 96$

$x = 2$

$y = 4$

20. Simplify: $\dfrac{6x^2 + 9x}{3x^2 - 12x}$

$$\frac{3x(2x+3)}{3x(x-4)}$$

$$\frac{2x+3}{x-4}$$

21. Construct a 2×2 matrix $A = [a_{ij}]$ whose elements are given by $a_{ij} = 2i - j$.

Solution: $a_{11}=1\ \ a_{12}=0\ \ a_{21}=3\ \ a_{22}=2$

$$A=\begin{bmatrix}1 & 0\\ 3 & 2\end{bmatrix}$$

22. Let $A=\begin{pmatrix}3 & 2\\ 5 & 1\end{pmatrix}$ and $B=\begin{pmatrix}8 & -1\\ 4 & 3\end{pmatrix}$. Find the matrix C, if C = 2A + B.

Solution: $C=2\begin{bmatrix}3 & 2\\ 5 & 1\end{bmatrix}+\begin{bmatrix}8 & -1\\ 4 & 3\end{bmatrix}$

$$C=\begin{bmatrix}14 & 3\\ 14 & 5\end{bmatrix}$$

23. Find the coordinates of the point which divides the line segment joining (–3, 5) and (4, –9) in the ratio 1 : 6 internally.

Solution: $P\left(\dfrac{lx_2+\mathrm{m}x_1}{l+\mathrm{m}},\dfrac{ly_2+\mathrm{m}y_1}{l+\mathrm{m}}\right)$

$$P\left(\frac{1\times 4+6\times -3}{1+6},\frac{1\times -9+6\times 5}{1+6}\right)$$

P(–2, 3)

24. "The points (0, a), a > 0 lie on x-axis for all a". Justify the truthness of the statement.

Solution: The statement is not true.

Reason: All the points are lie on the y-axis.

25. In ΔPQR, AB||QR. If AB is 3 cm, PB is 2 cm and PR is 6 cm, then find the length of QR.

Solution: $\dfrac{AB}{QR}=\dfrac{PB}{PR}$

$$\frac{3\text{ cm}}{QR}=\frac{2\text{ cm}}{6\text{ cm}}$$

$$QR=\frac{6\times 3}{2}=9\text{ cm}$$

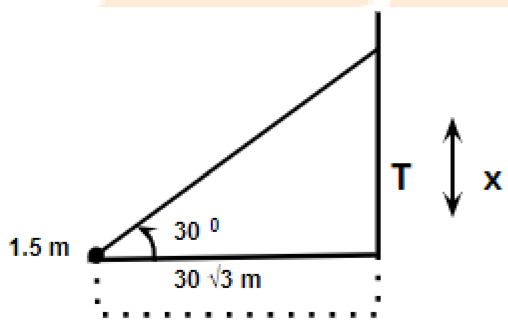
26. The angle of elevation of the top of a tower as seen by an observer is 30°. The observer is at a distance of $30\sqrt{3}$ m from the tower. If the eye level of the observer is 1.5 m above the ground level, then find the height of the tower.

Solution: CD = 30 m

$$\tan 30 = \frac{T}{30\sqrt{3}}$$

T = 30 m

Total length = 30 + 1.5 = 31.5 m



27. The total surface area of a solid right circular cylinder is 1540 cm^2. If the height is four times the radius of the base, then find the height of the cylinder.

Solution: $2\pi r\left(n+r\right)=1540$ n = 4r

$\boxed{h = 28cm}$

28. The smallest value of a collection of data is 12 and the range is 59. Find the largest value of the collection of data.

Solution:

Rang = L – S

59 = L – 12

L = 71

29. In tossing a fair coin twice, find the probability of getting:

(i) Two heads

(ii) Exactly one tail

Solution: Probability of getting too heads $P(B) = \frac{1}{4}$

Probability of getting exactly one tail = $P(B) = \frac{2}{4}$ or $\frac{1}{2}$

30. (a) If the volume of a solid sphere is $7241\frac{1}{7}$ cu.cm, then find its radius. $\left(\text{Take } \pi = \frac{22}{7}\right)$

OR

(b) If $x = a \sec \theta + b \tan \theta$ and $y = a \tan \theta + b \sec \theta$, then prove that $x^2 - y^2 = a^2 - b^2$.

Answer:

(a) $\frac{4}{3}\pi r^3 = 7241\frac{1}{7}$

$r^3 = 7241\frac{1}{7} \times \frac{3}{4} \times \frac{7}{22}$

$= 12$ cm

(b) $x^2 - y^2$

$= a^2 \sec^2 \theta + b^2 \tan^2 \theta + 2ab \sec \theta \tan \theta - a^2 \tan^2 \theta - b^2 \sec^2 \theta - 2ab \sec \theta \tan \theta$

$= a^2 - b^2$

31. Let A = {a, b, c, d, e, f, g, x, y, z}, B = {1, 2, c, d, e}and C = {d, e, f, g, 2, y}.

Verify $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$.

Solution:

$B \cup C = \{1, 2, c, d, e, f, g, y\}$

$A \setminus (B \cup C) = \{a, b, x, z\}$

$A \setminus B = \{a, b, f, g, x, y, z\}$

$A \setminus C = \{a, b, c, x, z\}$

$(A \setminus B) \cap (A \setminus C) = \{a, b, x, z\}$

32. Let A = {6, 9, 15, 18, 21}; B = {1, 2, 4, 5, 6} and f : A → B be defined by $f(x) = \frac{x-3}{3}$.

Represent f by:

(i) an arrow diagrams

(ii) a set of ordered pairs

(iii) a table

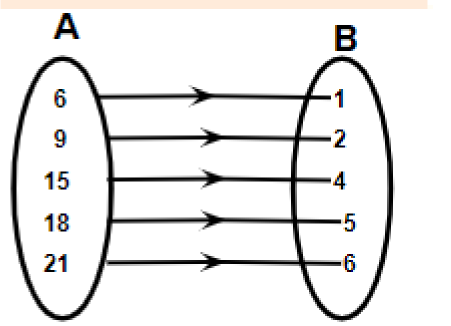
(iv) a graph

Solution:

$f(x) = \frac{x-3}{3}$

$f(6) = 1 \quad f(g) = 2, \quad f(15) = 4, \quad f(18) = 5, \quad f(21) = 6$

(i)

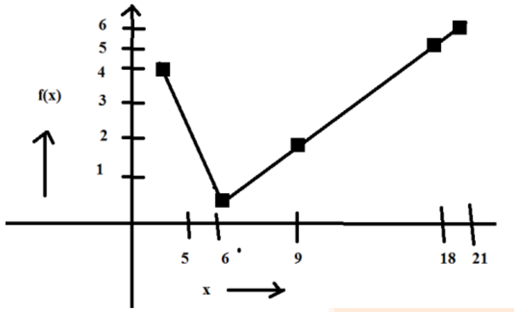


(ii) Ordered pairs $f = \{(6,1),(9,2),(15,4),(18,5),(21,6)\}$

(iii)

x	6	9	5	18	21
$f(x)$_	1	2	4	5	6

(iv)



33. Find the sum of the first 2n terms of the series $1^2 – 2^2 + 3^2 – 4^2 + ….$

Solution:

$1^2-2^2+3^2-4^2+.......$

$1-4+9-16......2n$ terms

$-3+(-7)+(-11)......$ upto n terms

$$s_n=\frac{n}{2}\left[2a+(n-1)d\right]$$

$$=\frac{n}{2}\left[2(-3)+(n-1)(-4)\right]$$

$= –n\ (2n + 1).$

34. Find the sum of n terms of the series 7 + 77 + 777 + ….

Solution:

$$s_n=7(1+11+111+..... \text{ upto n terms})$$

$$=\frac{7}{9}(9+99+999+...... \text{upto n terms})$$

$$=\frac{7}{9}\left[(10-1)+(100-1)+(1000-1)..... \text{ upto n terms}\right]$$

$$=\frac{7}{9}\left[10+10^2+10^3+..... \text{ upto n terms}\right]-n\Big]$$

$$=\frac{7}{9}\left[\frac{10(10^n-1)}{9}-n\right] \text{ or } \frac{70}{81}(10^n-1)-\frac{7}{9}n$$

35. The speed of a boat in still water is 15 km/hr. It goes 30 km upstream and return downstream to the original point in 4 hrs. 30 minutes. Find the speed of the stream.

Solution:

$$T_1 = \frac{30}{15+x}\text{hr}$$

$$T_2 = \frac{30}{15-x}\text{hr}$$

$$\frac{30}{15-x}+\frac{30}{15+x}=\frac{9}{2}$$

$$30(15+x)+30(15-x)=\frac{9}{2}\left(15^2-x^2\right)$$

$$2\left(450+\cancel{30}x+450-\cancel{30}x\right)=225\times 9-9x^2$$

$$\frac{1800}{9}=225-x^2$$

$225 – 200 = x^2$

$x = \pm 5$

speed of stream = ± 5 km/hr.

36. Find the values of a and b if $16x^4 – 24x^3 + (a – 1)x^2 + (b + 1)x + 49$ is a perfect square.

	4	–3	+7		
4	16 16	–24	(a – 1)	(b + 1)	49
8 – 3		–24 –24	a – 1 9		
8 – 6 + 7			a – 10 56	b + 1 –42	49 49
				0	
a = 66 b = –43	a = –46 b = 41			Taking –7 instead of +7	

37. $A = \begin{pmatrix} 5 & 2 \\ 7 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$ verify that $(AB)^T = B^TA^T$.

Solution:

$$AB = \begin{bmatrix} 8 & -3 \\ 11 & -4 \end{bmatrix}$$

$$AB^T = \begin{bmatrix} 8 & 11 \\ -3 & -4 \end{bmatrix}$$

$$B^T = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}$$

$$B^T = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 5 & 7 \\ 2 & 3 \end{bmatrix}$$

$$B^T A^T = \begin{bmatrix} 8 & 11 \\ -3 & -4 \end{bmatrix}$$

38. Find the area of the quadrilateral formed by the points (–4, –2), (–3, –5), (3, –2) and (2, 3).

Solution: Area of quadrilateral ABCD

$$= \frac{1}{2}\begin{Bmatrix} -4 & -3 & 3 & 2 & -4 \\ -2 & -5 & -2 & 3 & -2 \end{Bmatrix} s_9 \text{ units}$$

$$= \frac{1}{2}\{(+20+6+9-4)-(6-15-4-12)\}$$

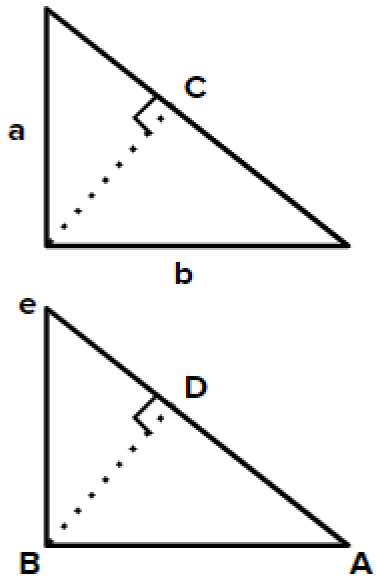
$$= \frac{1}{2}\times 56$$

= 28 sq units.

39. State and prove Pythagoras theorem.

Solution: Pythagoras theorem states that in right angled triangle, the square of a PWS the square of b is equal to squat of C where a, b, c are three division of triangle as shown.

$a^2 + b^2 = c^2$



Proof: construct ΔABC and $BD \perp AC$

We know $\Delta ADB \sim \Delta ABC$

$\therefore \frac{AD}{AB} = \frac{AB}{AC}$

or $AB^2 = AD \times AC$ …(1)

Also, $\Delta BOC \sim \Delta ABC$

$\frac{CD}{BC} = \frac{BC}{AC}$

$BC^2 = CD \times AC$ …(2)

Adding (1) and (2)

$AB^2 + BC^2 = AD \times AC + CD \times AC$

$= AC\,(AD + CD)$

$AB^2 + BC^2 = AC^2$ Proved

40. A flag post stands on the top of a building. From a point on the ground, the angles of elevation of the top and bottom of the flag post are 60° and 45° respectively. If the height of the flag post is 10 m, find the height of the building. $\left(\sqrt{3} = 1.732\right)$

Solution: $\tan 45 = \dfrac{BC}{AB}$

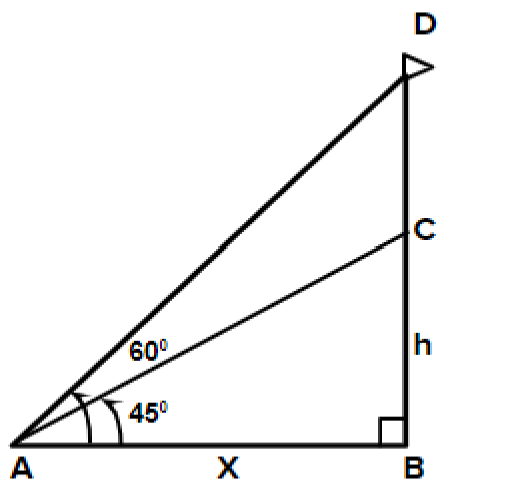
$\boxed{x = h}$

$\tan 60 = \dfrac{BD}{AB}$

$\sqrt{3} = \dfrac{h+10}{x} \qquad (\because h = x)$

$h = \dfrac{10\left(\sqrt{3}+1\right)}{2}$

$= 13.67 \text{ m}$



41. The perimeter of the ends of a frustum of a cone are 44 cm and 8.4 cm π cm. If the depth is 14 cm, then find its volume.

Solution:

R = 7 cm

r = 4.2 cm

vol. of frustum $= \dfrac{1}{3}\pi h\left(R^2 + r^2 + Rr\right)$

$= \dfrac{1}{3} \times \dfrac{22}{7} \times 14\left(7^2 + 4.2^2 + 7 \times 4.2\right)$

= 1408.58 cm^3 or 1408.6 cm^3

42. The length, breadth and height of a solid metallic cuboid are 44 cm, 21 cm and 12 cm respectively. It is melted and a solid cone is made out of it. If the height of the cone is 24 cm, then find the diameter of its base.

Solution: vol. of cone $= \frac{1}{2}\pi r^2 h$

vol. of cuboid = $l \times b \times h$

$= 44 \times 21 \times 12$ (melted)

Hence, vol. of cuboid = vol. of cone

$44 \times 21 \times 12 = = \frac{1}{3} \times \frac{22}{7} \times r^2 \times 24$

$$\frac{{}^{11}\cancel{22}\,\cancel{44} \times 21 \times \cancel{12} \times 3 \times 7}{\cancel{22}\,\cancel{2} \times \cancel{24}_{12}} = r^2$$

r ≈ 21 cm

dia = 2 × 21 = 42 cm

43. Find the coefficient of variation of the following data. 18, 20, 15, 12, 25

Solution: $\bar{x} = 18$

$\sum d = 0 \ \ \sum d^2 = 98$

$$\sigma = \sqrt{\frac{\sum d^2}{n}} = 4.428$$

$$C_V = \frac{\sigma}{x} \times 100$$

= 24.6

44. If a die is rolled twice, find the probability of getting an even number in the first time or a total of 8.

Solution:

n(s) = 36

$$P(A) = \frac{18}{36}$$

$$P(B) = \frac{5}{36}$$

$$P(A \cap B) = \frac{3}{36}$$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$$= \frac{18}{36} + \frac{5}{36} - \frac{3}{36}$$

$$= \frac{20}{36} \text{ or } \frac{5}{9}$$

45. (a) Find the GCD of the following polynomials $3x^4 + 6x^3 - 12x^2 - 24x$ and $4x^4 + 14x^3 + 8x^2 - 8x$.

OR

(b) A straight line cuts the coordinate axes at A and B. If the mid point of AB is (3, 2), then find the equation of AB.

Solution:

(a) Let

$$f(x) = 3x^4 + 6x^3 - 12x^2 - 24x$$
$$= 3x(x^3 + 2x^2 - 4x - 8)$$
$$g(x) = 4x^4 + 14x^3 + 8x^2 - 8x$$
$$= 2x(2x^3 + 7x^2 + 4x - 4)$$

$$
\begin{array}{r}
x^3 + 2x^2 - 4x - 8)2x^3 + 7x^2 + 4x - 4(2 \\
2x^3 + 4x^2 - 8x - 16 \\
- \quad - \quad + \quad + \\
\hline
3x^2 + 12x + 12 \\
3(x^2 + 4x + 4)
\end{array}
$$

The remainder si $3(x^2 + 4x + 4)$ is not equal to 0.

So, divide $x^3 + 2x^2 - 4x + 8$ by $x^2 + 4x + 4$

$$x^2+4x+4)x^3+2x^2-4x-8(x-2$$

$$x^3+4x^2+4x$$

$$- \quad - \quad + \quad +$$

$$-2x^2-8x-8$$

$$-2x^2-8x-8$$

$$0$$

Hence, GCD is x (x^2 + 4x + 4)

(b) Intersecting point on x axis = (a, 0)

Intersecting point on y axis = (0, b)

a = 6 b =4

$$\frac{x}{a}+\frac{y}{b}=1$$

2x + 3y – 12 = 0

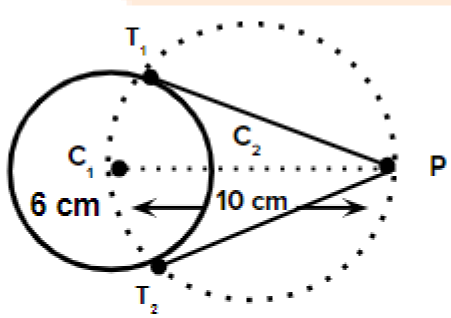
46.

(a) Draw the two tangents from a point which is 10 cm away from the centre of a circle of radius 6 cm. Also, measure the lengths of the tangents.

OR

(b) Construct a cyclic quadrilateral ABCD, given AB = 6 cm, ∠ABC =70°, BC = 5 cm and ∠ACD = 30°.

Solution:

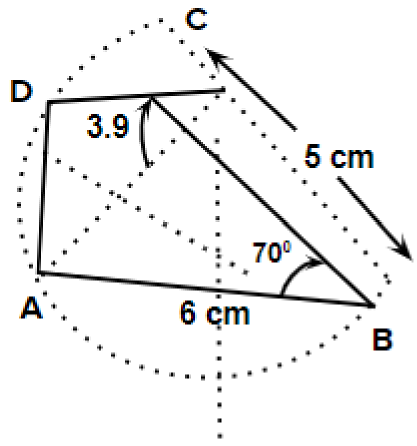


$T_1P = T_2P$ = 8 cm

(a) 1. Draw circle C_1 of 6 cm

2. Draw circle C_2 of 5 cm

3. Measure $T_1 P = T_2 P$ with scale.



(b) Draw AB = 6 cm

Draw = $\angle ABC = 70°$

@ $\angle ABC$ draw are of 5 cm complete ΔABC.

Draw $\perp$ bisectors of all three side and draw circumcircle. Draw $\angle ADC = 30°$ where ever line CD touch circle name it as D. Join ABCD